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Editor-in-Chief
*IEEE Transactions on Power Systems*

Dear Editor,

We submit for your consideration the manuscript titled:

**Field Validation of the Self-Adaptive Model-Based Protection System on 157 Production IBR Fault Recordings from a 33 kV DSO Network**

**Motivation.** Inverter-based resource (IBR) proliferation has profoundly altered fault-current signatures in distribution networks, yet most protection algorithms are validated exclusively on synthetic or hardware-in-the-loop data. The gap between simulation-domain performance and field performance is poorly characterised, particularly for high-impedance faults (HIF) and legacy non-compliant inverters.

**Contributions.** This paper makes five distinct contributions:

1. **Field dataset curation (C1).** We curate and label 157 disturbance-fault recordings from DSO-A, a real 33 kV network with 12 substations and 47 feeders, covering January 2023 to December 2025 (dataset protocol: TR-73).
2. **Synthetic-vs-real gap characterisation (C2).** Systematic comparison reveals a −17.6% HIF detection gap and +1.1 pp false-alarm increase when the SAMBPS framework is applied unmodified to field data.
3. **F1 correction: legacy-IBR $\kappa_n$-trajectory detector (C3).** A sliding-window $\kappa_n$ rise-rate flag identifies pre-IEEE-2800 inverter saturation and routes the estimator to a one-cycle hold, recovering zone accuracy by +3.5 pp.
4. **F2 correction: split-window HIF estimator (C4).** A two-regime negative-sequence estimator handles evolving HIF-to-bolted transitions, improving HIF detection by +7.4 pp.
5. **F3 correction: shape-factor calibration (C5).** A scalar $\beta$ factor ($\beta_{\text{tri}} = 8/\pi^2 \approx 0.811$) corrects the asymmetric limiter waveform for unusual inverter profiles, closing the remaining detection gap.

Combined, the three corrections (F1–F3) raise zone accuracy from 89.2% to 95.5% and reduce false alarms from 2.1% to 1.4%, approaching the synthetic benchmark of 98.2% and 1.0% respectively.

**Fit to the journal.** *IEEE Transactions on Power Systems* is the premier venue for rigorous field-validated protection studies. This work provides the first large-scale IBR protection field-validation dataset in a real DSO context, directly advancing the journal's stated interest in grid-edge protection and IBR integration.

**Suggested reviewers.**

- Prof. Ali Hooshyar (University of Toronto) — IBR protection
- Prof. Geza Joos (McGill University) — microgrid and distribution protection

- Prof. Wilsun Xu (University of Alberta) — power quality and HIF detection
- Dr. Evangelos Farantatos (EPRI) — field validation of protection algorithms

This manuscript is not under review elsewhere. All authors have approved the submission. The field dataset was provided under a research agreement with DSO-A; identifying substation details have been anonymised.

Sincerely,

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